

Muhammad Mahdi Qasimi

Doctoral Researcher, Faculty Engineering and
Natural Sciences, Tampere University

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RESEARCH FIELDS

Cold spray coatings, coating characterization, confined space deposition, particle impact behavior and process optimization.

EMPLOYMENT

- 2026 – Tampere University, Materials Characterization/Coating Technologies Group –
Doctoral Researcher
- 2025 Tampere University, Coating Technologies Group – Research assistant

EDUCATION

- 2026 – Tampere University
Doctor of Science in Materials Science and Engineering
- 2023 – 2025 Tampere University
Master of Science in Materials Science and Engineering, with distinction
- 2018 – 2022 University of the Punjab
Bachelor of Engineering in Metallurgy and Materials Engineering

RESEARCH OUTPUT

Investigation of In-flight Behavior and Coating Property Correlation in Internal Diameter Cold Spray – Manuscript (not published)

In-flight Behavior and Coating Property Correlation for IDSC, [link](#)
International Cold Spray Conference & Expo 2026, Chennai, India – Conference Talk

Development of data-driven predictive tool for coating quality assessment, as an experimental tool for analyzing particle in-flight behavior spray and its correlation with coating properties in cold spray – Developed Tool

Estimation of Deformation in Dome-shape Structure Made of FGM for Aerospace Applications
All Pakistan Materials Engineering Competition 2022, UET Lahore, Pakistan – Poster

EDUCATION

2022	Winner Comp 1.0, All Pakistan Materials Engineering Competition (simulation)
2022	Best Poster Award Final year research project, University of the Punjab,
2021	Competition Award NACE Pakistan

TECHNICAL SKILLS

Cold Spray & Diagnostics — Cold spray system operation; particle velocity measurement; camera calibration

Materials Characterization & Testing — SEM; Optical microscopy; hardness testing; metallographic sample preparation; corrosion and electrochemical testing (EIS)

Computational — Python (data analysis, scikit-learn); ANSYS (finite element modelling); SolidWorks; OriginLab

Research Methods — Experimental design; data analysis and interpretation; scientific writing and reporting

MOOCs & TRAINING

MOOCs — Machine Learning Specialization, Coursera (in progress); Introduction to High-Throughput Materials Development, Coursera; Materials Data Science and Informatics, Coursera

NDT Training — Introductory training in VT, MT, PT and UT, supervised by Level III NDT personnel

SUPERVISORS

Prof. Minnamari Vippola — minnamari.vippola@tuni.fi

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